



# JSON

This is a quick reference cheat sheet for understanding and writing JSON format configuration files.

## # Getting Started

### Introduction

JSON is a lightweight text-based open standard designed for human-readable data interchange.

- JSON stands for JavaScript Object Notation
- JSON is easy to read and write.
- JSON is language agnostic data-interchange format
- JSON filename extension is `.json`
- JSON Internet Media type is `application/json`

### Examples

```
{  
  "name": "Jason",
```

```

    "age": 39,
    "height": 1.92,
    "gender": "M",
    "salary": 70000,
    "married": true,
    "children": [
      {"name": "Tom", "age": 9, "gender": "M"},
      {"name": "Ava", "age": 7, "gender": "F"}
    ]
  }

```

#### Types

|         |                                         |
|---------|-----------------------------------------|
| Number  | Double precision floating-point         |
| String  | Series of characters                    |
| Boolean | true or false                           |
| Array   | Ordered sequence of values              |
| Value   | String, Number, Boolean, null etc       |
| Object  | Unordered collection of key/value pairs |
| null    | Null or Empty                           |

#### String

|    |               |
|----|---------------|
| \" | Double quote  |
| \\ | Backslash     |
| \/ | Forward slash |
| \b | Backspace     |
| \f | Form feed     |
| \n | Newline       |

|                 |                            |
|-----------------|----------------------------|
| <code>\r</code> | Carriage return            |
| <code>\t</code> | Tab                        |
| <code>\u</code> | Trailed by four hex digits |

### Examples

```
{
  "url": "https://quickref.me",
  "msg" : "Hi,\n\"QuickRef.ME\"",
  "intro": "Share quick reference and cheat sheet for developers."
}
```

### Invalid String

```
{ "foo": 'bar' }
```

Have to be delimited by double quotes

### Number

|          |                                        |
|----------|----------------------------------------|
| Integer  | Digits 1-9, 0 and positive or negative |
| Fraction | Fractions like 0.3, 3.9                |
| Exponent | Exponent like e, e+, e-, E, E+, E      |

### Examples

```
{
  "positive" : 12,
  "negative" : -1,
  "fraction" : 10.25,
  "exponent" : 1.0E+2,
  "zero" : 0
}
```

### Invalid Number

```
{ "foo": 0xFF }
```

## Objects

```
{  
  "color": "Purple",  
  "id": "210",  
  "composition": {  
    "R": 70,  
    "G": 39,  
    "B": 89  
  },  
  "empty_object": {}  
}
```

Multiple key/value pairs separated by a comma

## Arrays

```
[1, 2, 3, 4, 5]
```

Begins with [ and ends with ]

## Array of objects

```
{  
  "children": [  
    {"name": "Jimmy Smith", "age": 15},  
    {"name": "Sammy Sosa", "age": 12}  
  ]  
}
```

## Object of arrays

```
{
  "attributes": ["a1", "a2"],
  "methods": ["getter", "setter"],
}
```

2D Array

```
{
  "my_sequences": [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9, 0],
    [10, 11]
  ]
}
```

Object of objects

```
{
  "Mark McGwire": {
    "hr": 65,
    "avg": 0.278
  },
  "Sammy Sosa": {
    "hr": 63,
    "avg": 0.288
  }
}
```

Nested

```
{
  "Jack": {
    "id": 1,
    "name": "Franc",
    "salary": 25000,
    "hobby": ["a", "b"],
    "location": {

```

```
    "country": "A", "city": "A-A"
  }
}
```

## # Access JSON in JavaScript

### Access Object

```
let myObject = {
  "name": "Jason",
  "last": "Doe",
  "age": 39,
  "gender": "M",
  "salary": 70000,
  "married": true
};
```

|                               |           |
|-------------------------------|-----------|
| <code>myObject.name</code>    | "Jason"   |
| <code>myObject["name"]</code> | "Jason"   |
| <code>myObject.age</code>     | 39        |
| <code>myObject.other</code>   | undefined |
| <code>myObject[0]</code>      | undefined |

### Access Nested

```
let myObject = {
  "ref": {
    "name": 0,
    "last": 1,
```

```

    "age": 2,
    "gender": 3,
    "salary": 4,
    "married": 5
  },
  "jdoe": [
    "Jason",
    "Doe",
    39,
    "M",
    70000,
    true
  ],
  "jsmith": [
    "Tom",
    "Smith",
    42,
    "F",
    80000,
    true
  ]
];

```

|                                     |                          |
|-------------------------------------|--------------------------|
| <code>myObject.ref.age</code>       | 2                        |
| <code>myObject["ref"]["age"]</code> | 2                        |
| <code>myObject.jdoe</code>          | ["Jason", "Doe", 39 ...] |
| <code>myObject.jsmith[3]</code>     | "F"                      |
| <code>myObject[1]</code>            | undefined                |

Access Array of Objects

```

let myArray = [
  {
    "name": "Jason",
    "last": "Doe",

```

```

    "age": 39,
    "gender": "M",
    "salary": 70000,
    "married": true
  },
  {
    "name": "Tom",
    "last": "Smith",
    "age": 42,
    "gender": "F",
    "salary": 80000,
    "married": true
  },
  {
    "name": "Amy",
    "last": "Burnquist",
    "age": 29,
    "gender": "F",
    "salary": 60000,
    "married": false
  }
];

```

|            |                          |
|------------|--------------------------|
| myArray[0] | { "name": "Jason", ... } |
|------------|--------------------------|

|                 |       |
|-----------------|-------|
| myArray[1].name | "Tom" |
|-----------------|-------|

|               |    |
|---------------|----|
| myArray[1][2] | 42 |
|---------------|----|

|            |           |
|------------|-----------|
| myArray[3] | undefined |
|------------|-----------|

|                   |                           |
|-------------------|---------------------------|
| myArray[3].gender | TypeError: Cannot read... |
|-------------------|---------------------------|

Access Array

```

let myArray = [
  "Jason",
  "Doe",

```



```
39,  
"M",  
70000,  
true  
];
```

|            |           |
|------------|-----------|
| myArray[1] | "Doe"     |
| myArray[5] | true      |
| myArray[6] | undefined |

## # Also see

[JSON](#) ([json.org](#))

[JSON Editor Online](#) ([jsoneditoronline.org](#))

[Convert JSON Array to Markdown Table, CSV and more](#) ([tableconvert.com](#))

## Related Cheatsheet

**ES6 Cheatsheet**

Quick Reference

**Express Cheatsheet**

Quick Reference

**Kubernetes Cheatsheet**

Quick Reference

**TOML Cheatsheet**

Quick Reference

## Recent Cheatsheet

**InfrontJS Cheatsheet**

Quick Reference

**Remote Work Revolution Cheatsheet**

Quick Reference

**Homebrew Cheatsheet**

Quick Reference

**PyTorch Cheatsheet**

Quick Reference

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